Example Co.

Type 1 Gage Study

27 Oct 2026

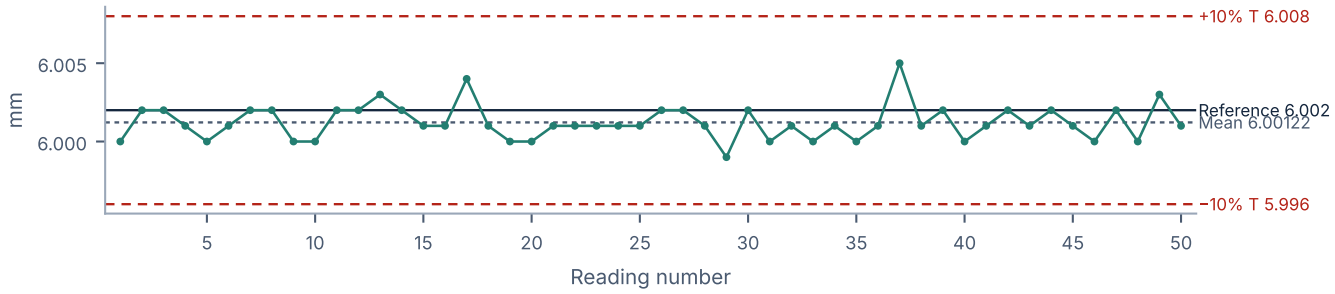
PART NUMBER **SAMPLE-003**	CHARACTERISTIC **Outer diameter (simulated data)**	SPECIFICATION **LSL 5.97 · USL 6.03 mm**
GAGE **Example gage**	REFERENCE PART **x_m = 6.002 mm**	CUSTOMER Not given

ACCEPTABLE

Cg is 1.80, at or above the 1.33 minimum. Cgk is 1.56, at or above the 1.33 minimum. Resolution is 1.67% of the tolerance, within the 5% limit. The bias of -0.0008 mm is statistically significant (p < 0.001); AIAG MSA recommends correcting it.

CG	CGK	BIAS	RESOLUTION
1.80	**1.56**	**-0.0008**	**1.7%**
Repeatability · minimum 1.33	Repeatability and bias · minimum 1.33	Significant	Of tolerance · limit 5%

Readings on the reference part



Results	
Mean ($\bar{x}$)	**6.00122 mm**
Standard deviation (s)	**0.001112 mm**
6s	**0.006671 mm**
Bias ($\bar{x} - x_m$)	**-0.00078 mm**
|Bias| / s vs. criterion	**0.702 vs. 0.284**
Bias t-test	**t = -4.96, p < 0.001**
Cg 95% interval 1.44 to 2.15	**1.80**
Cgk 95% interval 1.24 to 1.89	**1.56**

Study	
Reference value (x_m)	**6.002 mm**
Tolerance	**0.06000 mm (T = USL - LSL)**
Readings	**50**
Gage resolution	**0.001 mm (1.67% of T)**

Smallest workable tolerance (T_{min})	
For Cg ≥ 1.33	**0.04436 mm**
For Cgk ≥ 1.33	**0.05216 mm**
For resolution ≤ 5%	**0.02000 mm**

Prepared by · Sample report

Approved by

Date

Method and decision rules

How every number in this report was made

Method

Type 1 gage study (Procedure 1, Robert Bosch GmbH, Booklet No. 10, Capability of Measurement and Test Processes). One calibrated reference part with reference value x_m is measured 50 times, removed and re-clamped between readings.

$$Cg = 0.2\,T / 6s$$
$$Cgk = (0.1\,T - |\bar{x} - x_m|) / 3s$$

T is the tolerance (T = USL - LSL) and s the sample standard deviation (n − 1). 95% intervals: chi-square interval for Cg and the normal approximation for Cgk (Montgomery, Introduction to Statistical Quality Control, 6th ed., equations 8.20 and 8.21).

Bias test: $|\bar{x} - x_m| / s$ compared with $t_{n-1;\,1-\alpha/2} / \sqrt{n}$ (one-sample t-test, α = 0.05; AIAG MSA, Booklet No. 10 Appendix C)

$T_{min}(Cg) = 1.33 \times 6s / 0.2$ $T_{min}(Cgk) = (1.33 \times 3s + |\bar{x} - x_m|) / 0.1$

Decision rules

- **Cg and Cgk both 1.33 or more**: capable.
- **Either below 1.33**: not capable.
- **Resolution over 5% of the tolerance**: acceptable only if no gage with finer resolution is available; over 10% is not acceptable.

A significant bias doesn't change the verdict here, but AIAG MSA recommends correcting it. Your customer's requirements apply if they differ.

Receipt

Method	type 1 gage study (Cg, Cgk) (Bosch Booklet No. 10, Procedure 1; bias t-test per AIAG MSA)
Sample	50 readings
Reference value	6.002
LSL	5.97
USL	6.03
Tolerance	T = USL - LSL
Required Cg and Cgk	1.33
Alpha	0.05
Resolution	0.001
Cg interval	chi-square (Montgomery eq. 8.20)
Cgk interval	normal approximation (Montgomery eq. 8.21)
Bias test	one-sample t-test, two-sided
Constant: Cg tolerance share	0.2
Constant: Cgk tolerance share	0.1
Engine	Mensuro engine 0.1.0
Generated	2026-10-27

- The bias of -0.00078 is statistically significant (p < 0.001). AIAG MSA recommends correcting it, for example by adjusting the gage, or correcting each reading.

How the math is checked

Before release, Mensuro's Type 1 engine must reproduce Bosch Booklet No. 10's worked examples (two-sided and one-sided) and its bias criterion table, and the interval formulas must reproduce Montgomery's Examples 8.4 and 8.5. The t-test comes from SciPy. These checks rerun on every change.

Data

All 50 readings, in mm, in the order measured

Readings	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9
1–10	6.000	6.002	6.002	6.001	6.000	6.001	6.002	6.002	6.000	6.000
11–20	6.002	6.002	6.003	6.002	6.001	6.001	6.004	6.001	6.000	6.000
21–30	6.001	6.001	6.001	6.001	6.001	6.002	6.002	6.001	5.999	6.002
31–40	6.000	6.001	6.000	6.001	6.000	6.001	6.005	6.001	6.002	6.000
41–50	6.001	6.002	6.001	6.002	6.001	6.000	6.002	6.000	6.003	6.001